**NAME OF SCHOOL: MATER MISERICORDIAE SECONDARY SCHOOL, RUMOMASI**

TERM: FIRST 2026/2027 ACADEMIC SESSION

WEEK: ONE

CLASS: SS1 SUBJECT: Biology

# TOPIC: Characteristics of Living Things

## Introduction

- Living things are organisms that carry out the activities of life — movement, respiration, sensitivity, growth, reproduction, excretion and nutrition. Biologists use the acronym MRS GREN to remember these life processes.

## Key life processes

- **Movement** — living things change position or move parts of themselves (plants move towards light).
- **Respiration** — the breakdown of food to release energy (glucose + oxygen → carbon dioxide + water + energy).
- **Sensitivity (irritability)** — the ability to detect and respond to changes in the environment, e.g. a flower tracking the sun.
- **Growth** — a permanent increase in size and dry mass, brought about by cell division and enlargement.
- **Reproduction** — the production of new individuals so the species continues.
- **Excretion** — the removal of metabolic waste, e.g. carbon dioxide, urea and excess water.
- **Nutrition** — the taking in and use of food (plants make their own; animals feed on other organisms).

## Dormancy and hibernation

- Dormant seeds do not show visible life processes but resume activity when conditions (water, warmth, oxygen) return — they are still living. This is why 'is it alive?' cannot be answered by watching for a few minutes alone.

## CLASS EVALUATION

1. Name the seven life processes in the MRS GREN order.
2. Distinguish between movement in a plant and movement in an animal.
3. Why is excretion different from egestion?
4. Give two reasons a dry seed is still described as a living thing.

**NAME OF SCHOOL: MATER MISERICORDIAE SECONDARY SCHOOL, RUMOMASI**

TERM: FIRST 2026/2027 ACADEMIC SESSION

WEEK: TWO

CLASS: SS1 SUBJECT: Biology

# TOPIC: Classification of Living Things

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## Introduction

- Classification is the grouping of organisms on the basis of shared features. It makes the study of the millions of organisms on earth orderly and shows their relationships.

## The five kingdoms

- **Monera** — bacteria: single-celled, no true nucleus (prokaryotic).
- **Protista** — amoeba, paramecium, euglena: single-celled with a true nucleus.
- **Fungi** — moulds, mushrooms, yeast: saprophytic, no chlorophyll, cell wall of chitin.
- **Plantae** — green plants: photosynthetic, cell walls of cellulose, mostly fixed to the ground.
- **Animalia** — animals: heterotrophic, move, no cell wall, store food as glycogen.

## Why classification matters

- It reveals evolutionary relationships, allows scientists worldwide to use the same names (binomial nomenclature by Carolus Linnaeus), and helps identify harmful or useful species quickly. Every organism has a genus and a species name, e.g. *Homo sapiens*.

## CLASS EVALUATION

1. List the five kingdoms with one example each.
2. Which kingdom has organisms without a true nucleus?
3. Who developed the binomial naming system and what does it consist of?
4. State two reasons for classifying organisms.

**NAME OF SCHOOL: MATER MISERICORDIAE SECONDARY SCHOOL, RUMOMASI**

TERM: FIRST 2026/2027 ACADEMIC SESSION

WEEK: THREE

CLASS: SS1 SUBJECT: Biology

# TOPIC: Cell Structure & Functions

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## Introduction

- The cell is the basic unit of life. All living things are made of cells; new cells arise from existing cells. A cell consists of a membrane, cytoplasm and a nucleus (in eukaryotic cells).

## Cell organelles

- **Nucleus** — controls all cell activities and carries genetic material (DNA).
- **Cell membrane** — controls what enters and leaves the cell (semi-permeable).
- **Mitochondria** — site of respiration; the power house that releases energy.
- **Ribosomes** — protein synthesis.
- **Cytoplasm** — the jelly-like medium where chemical reactions take place.
- **Vacuole** — stores cell sap (large in plant cells).
- **Chloroplasts** — contain chlorophyll for photosynthesis (plant cells only).
- **Cell wall** — rigid cellulose wall that gives a plant cell its shape and protection.

## Plant vs animal cells

- Plant cells have a cellulose cell wall, chloroplasts and one large central vacuole; animal cells have none of these (or only small temporary vacuoles). Both have a nucleus, membrane, cytoplasm, mitochondria and ribosomes.

## CLASS EVALUATION

1. Draw and label a typical plant cell.
2. Which organelle is the site of respiration?
3. State three differences between plant and animal cells.
4. What is the function of the cell membrane?

**NAME OF SCHOOL: MATER MISERICORDIAE SECONDARY SCHOOL, RUMOMASI**

TERM: FIRST 2026/2027 ACADEMIC SESSION

WEEK: FOUR

CLASS: SS1 SUBJECT: Biology

# TOPIC: Cell Division (Mitosis)

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## Introduction

- Mitosis is the division of a cell into two identical daughter cells with the same number of chromosomes. It produces growth, repair of worn-out tissues and asexual reproduction.

## Stages of mitosis

- **Prophase** — chromosomes become visible, each made of two chromatids joined at the centromere; the nuclear membrane breaks down.
- **Metaphase** — chromosomes line up at the equator of the spindle.
- **Anaphase** — chromatids separate and move to opposite poles.
- **Telophase** — two new nuclei form and the cytoplasm divides (cytokinesis), giving two identical cells.

## Importance

- It maintains the chromosome number (diploid $\rightarrow$ diploid), allows a single fertilised egg to become millions of cells, and heals wounds. Uncontrolled mitosis can produce tumours.

## CLASS EVALUATION

1. Define mitosis.
2. Arrange the four stages of mitosis in order and describe one event of each.
3. Why are the daughter cells identical?
4. State two importance of mitosis in living organisms.

**NAME OF SCHOOL: MATER MISERICORDIAE SECONDARY SCHOOL, RUMOMASI**

TERM: FIRST 2026/2027 ACADEMIC SESSION

WEEK: FIVE

CLASS: SS1 SUBJECT: Biology

# TOPIC: Plant Tissues

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## Introduction

- A tissue is a group of similar cells working together to perform one function. In plants, tissues are grouped into meristematic (dividing) and permanent tissues.

## Meristematic tissues

- Found at the tips of roots and shoots (apical meristems) and in the cambium; cells divide continuously by mitosis and are thin-walled with dense cytoplasm, giving the plant new growth.

## Permanent tissues

- **Parenchyma** — thin-walled living cells for storage and photosynthesis.
- **Collenchyma** — thickened living cells giving young stems support.
- **Sclerenchyma** — thick, lignified dead cells for strong support.
- **Xylem** — transports water and mineral salts upward and supports the plant (tracheids and vessels).
- **Phloem** — transports food (sucrose) from leaves to all parts of the plant through sieve tubes and companion cells.

## CLASS EVALUATION

1. Define a tissue.
2. Distinguish between xylem and phloem.
3. Where are meristematic tissues found?
4. Name the supporting tissues in plants and state how they differ.

**NAME OF SCHOOL: MATER MISERICORDIAE SECONDARY SCHOOL, RUMOMASI**

TERM: FIRST 2026/2027 ACADEMIC SESSION

WEEK: SIX

CLASS: SS1 SUBJECT: Biology

# TOPIC: Animal Tissues & Organisation

## Introduction

- In animals, four main tissue types build every organ: epithelial, connective, muscular and nervous tissues.

## The four tissue types

- **Epithelial** — covers surfaces and lines organs (e.g. skin, lining of the mouth); protects, absorbs and secretes.
- **Connective** — supports and binds (blood, bone, cartilage, tendons and adipose tissue).
- **Muscular** — generates movement (voluntary/skeletal, involuntary/smooth and cardiac muscle).
- **Nervous** — receives and transmits impulses (neurons make up the brain, spinal cord and nerves).

## Levels of organisation

- Cells → tissues → organs (e.g. the heart) → systems (e.g. circulatory system) → organism. This hierarchy shows how the body of a mammal is organised from the simplest unit upwards.

## CLASS EVALUATION

1. Name the four animal tissue types with one function of each.
2. Which tissue transmits nerve impulses?
3. Give the correct order: organ → cell → system → tissue.
4. What tissue types make up the heart?